Alberto Marrone

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Summary

M.Sc. Electronics Engineering student at Politecnico di Milano focused on board-level hardware design, PCB layout, and power electronics. Designed and validated a 6-layer, 90 Ω controlled-impedance USB 3.2 hub from schematic through fabrication and bring-up, achieving 102 MB/s SuperSpeed throughput on measured hardware. Designed a 400W, 240 kHz GaN active-clamp flyback, achieving 97.6% simulated peak efficiency and a 6× reduction in simulated switching losses through device-level optimization.

Experience

Sept 2024 – Feb 2025 — **Bachelor's Thesis Project**, University of Bologna

Led end-to-end hardware development of a 240 kHz GaN-based active-clamp flyback microinverter (400W). Achieved a 6× reduction in **simulated** primary switching losses (0.12W to 0.02W) through device-level optimization, with 97.6% simulated peak efficiency at 160W and 92.3% at 400W via magnetic modeling (ETD34) and high-frequency layout refinement. Designed a 4-layer PCB in KiCad with minimized power loops, controlled return paths, EMI-aware routing, and isolated sensing (AMC0311S). Refined manufacturer SPICE models to improve transient simulation accuracy.

Feb 2024 – Jul 2024 — **Research Intern – Power Electronics**, University of Bologna

Designed and analytically modeled a 400W resonant LLC converter, including resonant tank design, operating-region analysis, and system-level loss evaluation. Built a reusable C/MATLAB sizing tool implementing Q–m gain curves and ZVS/ZCS boundary analysis to accelerate converter design iterations and eliminate manual calculation errors. Achieved up to 97.9% **simulated** peak efficiency across a 100 kHz–1 MHz operating range.

Selected Hardware Projects

Mar 2026 – Jun 2026 — **USB 3.2 Gen 1 4-Port Hub**, Independent Project

Designed a fully bus-powered USB 3.2 Gen 1 (5 Gbps) hub around the TI TUSB8044A featuring hardware-only USB-C attach detection, cold-socket VBUS control, and per-port current limiting without firmware. Engineered a 6-layer impedance-controlled PCB with 90 Ω differential routing and ≤0.15 mm intra-pair skew; manufacturer DFM review identified and corrected a via-in-pad issue before fabrication. Validated the fabricated board end-to-end: all power rails within 1% of target, USB interfaces enumerating on first power-up, and 102 MB/s SuperSpeed throughput confirmed through functional testing. GitHub Repository

Feb 2026 – Jun 2026 — **STM32F746 Application Board**, UPV Coursework – 3-Person Team

Led hierarchical schematic capture and 4-layer impedance-controlled PCB layout (90 Ω USB, 100 Ω Ethernet, 120 Ω CAN) for a custom STM32F746ZG application board integrating Ethernet (LAN8742A PHY), dual CAN, SPI flash, and analog front-end interfaces; collaborated with two teammates in real time via Altium 365. Design complete through manufacturing outputs (Gerbers, BOM, pick-and-place); not yet fabricated. GitHub Repository

Apr 2026 – Present — **DRSSTC Fiber-Optic Interrupter**, Independent Project

Designed and hand-assembled a galvanically isolated interrupter control board (STM32F405) for a Tesla coil driver, implementing a dual-redundant hardware safety chain with PMOS power cutoff independent of firmware state and hardware PWM shutdown. Routed a 4-layer controlled-impedance PCB (90 Ω USB, 50 Ω FSMC/SDIO). Verified power rails, MCU programming, and USB communication during initial bring-up; diagnosed and repaired PCB trace damage during connector rework. Final bring-up and validation ongoing. GitHub Repository

Education

Feb 2025 – Present	**Politecnico di Milano**, Master's Degree in Electronics Engineering *Relevant Coursework*: Digital Integrated Circuit Design, Analog Circuit Design, Hardware Architectures for Embedded and Edge AI, Design of Hardware Accelerators.
Feb 2026 – Jun 2026	**Universitat Politècnica de València**, Erasmus Exchange Program (Semester 2 of PoliMi Master's) *Relevant Coursework*: Embedded Systems for IoT, Electric Motors Efficiency, Electric Vehicles, PCB Design.
Sep 2021 – Feb 2025	**University of Bologna**, Bachelor's Degree in Electronics Engineering *Relevant Coursework*: Power Electronics, Electronic Measurements, Computer Architectures.

Skills

Hardware & Power Systems: Board-level power architecture, GaN/SiC converters, DC-DC/DC-AC topologies, resonant converters, PDN design, feedback control, loss modeling, efficiency optimization.
PCB Design: KiCad, Altium Designer, Cadence Allegro/OrCAD; multilayer PCB layout, high-speed digital routing, impedance control, return-path control, EMI-aware layout, DFM.
Signal & Power Integrity: Parasitic modeling, switching transient analysis, snubber optimization, SPICE-to-hardware correlation, Saturn PCB Toolkit.
Simulation: LTspice, MATLAB, PSpice, COMSOL Multiphysics, CST Studio Suite, AWR.
Programming & Interfaces: C, Python, MATLAB scripting, Git/GitHub; USB, Ethernet/RMII, CAN, SPI, I^2C, UART, FSMC/SDIO.
Laboratory: Oscilloscopes, spectrum analyzers, nanoVNA, programmable power supplies, electronic loads.

Certifications

Cadence Student Ambassador | Cadence Allegro X Advanced Package Designer | Altium Education Global Scholar | Altium PCB Design Essentials Accredited

Languages

Italian – Native | **English** – Full Professional | **Spanish** – Basic